

Shelton Bai

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EDUCATION

University of Pittsburgh

Bachelor of Science in Computer Science

Pittsburgh, PA

Sep. 2020 – Dec 2023

EXPERIENCE

Software Developer

June 2024 – Present

CAO Group, Inc.

West Jordan, UT

- Developed an AI chatbot for dentists to source information from academic papers using retrieval augmented generation
- Fixed issues with AT commands setting incorrect values in Bluetooth Low Energy modules

Software Development Intern

May 2023 – August 2023

CAO Group, Inc.

West Jordan, UT

- Developed iOS applications in Objective-C to be paired with Bluetooth dental curing lasers, replacing the older Qt touchscreens for a modernized user experience
- Centralized management of laser presets and configurations, simplifying updates to existing laser firmware
- Created a Python script to automate address validation of dental offices for the sales and marketing departments, reducing manual workload by 30%
- Leveraged AWS (EC2, S3, RDS, and Lambda) to build a proof-of-concept cloud based file sharing system, to be used with a new digital dental headset

Software Development Intern

June 2021 – August 2021

CAO Group, Inc.

West Jordan, UT

- Wrote and tested embedded C for dental lasers and the Bluetooth paddle controllers for them
- Fixed a memory leak from a buffer which caused the lasers to malfunction after firing continuously for around 30 seconds
- Used C++ and Qt to help develop the user interface for the dental lasers

PROJECTS

Warforged DM | *Python, LangChain, LangGraph, Chroma, MongoDB, Docker*

- github.com/Shelton-Bai/warforged
- Personal webpage built using React, Spring Boot, and MySQL, containerized using Docker and run on an Oracle Cloud instance.
- Uses a Spring Web API to allow for quick and easy updates to content without having to rebuild the docker images

P2 Teambuilder | *Javascript, Java, React, TailwindCSS, Spring Boot, MySQL, Docker*

- github.com/Shelton-Bai/P2Teambuilder
- Full stack web application for creating and saving competitive Pokemon teams
- Sources information from Pokemon Showdown! and PokeAPI for up to date Pokemon data

SKILLS

Languages: Java, Python, C, C++, JavaScript, Typescript, C#, Objective-C, R, HTML/CSS

Frameworks & Libraries: React.js, React Native, Node.js, LangChain, LangGraph, Flask, PyQt, Qt, Spring Boot, TailwindCSS, ROS2

Developer Tools: Git, Docker, AWS, Oracle Cloud, VS Code, IntelliJ, PyCharm

Database Skills: PostgreSQL, MySQL, MongoDB, Chroma

CERTIFICATIONS

Oracle Certified Associate, Java SE 8 Programmer

AWS Certified Solutions Architect – Associate

February 2023 – Present

August 2023 – August 2026